Summer Delta Attention v5-Gated versus Kimi Delta Attention

Controlled Synthetic Associative-Recall Comparison and Memory–Accuracy Trade-offs

summer

Research manuscript – 23 October 2026 – Unreviewed preprint

Abstract

We compare Summer Delta Attention v5-Gated (SDA-5), a no-decay delta-rule state augmented with confidence-gated key–value (KV) retrieval, against the recurrent update underlying Kimi Delta Attention (KDA). A controlled NumPy benchmark uses $d_k = 32$, $d_v = 16$, 128 queries, 16 held-out seeds, sequence lengths 128–2,048, and independently tuned development hyperparameters. For 512 random KV pairs and exact-key queries, SDA-5 retaining 50% of the KVs achieves mean squared error (MSE) 0.4721 ± 0.0099 (mean $\pm$ standard error), compared with 0.9336 ± 0.0098 for a fixed channel-gated KDA recurrence proxy. However, the SDA-5 state and archive consume 100 KiB, whereas KDA uses 4 KiB, a 25-fold disparity. A direct indexed KV baseline yields MSE 0.4962 ± 0.0101 with 96 KiB. The KDA comparison implements the published recurrence but *does not train input-conditioned gates or reproduce the complete Kimi Linear model*; these results cannot establish superiority over KDA in language modeling. We quantify performance under query noise and longer histories, provide an algebraic characterization of SDA-5's correction, and release deterministic reproduction code. This study establishes a controlled proof of concept and identifies, rather than resolves, the remaining fair-comparison requirements.

Keywords: delta-rule memory; Kimi Delta Attention; indexed KV retrieval; associative recall; memory efficiency; reproducibility.

1 Introduction

Full attention uses content-dependent access to historical KV pairs but has cache storage growing with sequence length [1]. Delta-rule fast weights compress past associations into a fixed-size recurrent matrix [2], making writes efficient but vulnerable to interference. Kimi Delta Attention (KDA) builds on Gated DeltaNet by introducing channel-specific decay and efficient chunkwise computation [4, 3]. Kimi Linear further combines KDA with global multi-head latent attention (MLA); results from that full model should not be conflated with results from a single recurrent cell.

Summer Delta Attention v5-Gated (SDA-5) combines a no-decay delta state with a budgeted archive of explicit KV pairs. The present question is whether this hybrid produces more accurate associative recall than the KDA recurrent mechanism, and at what memory cost. Our contributions are: (i) a recurrence-level comparative implementation with explicit assumptions, (ii) development/test separation for hyperparameter selection, (iii) a memory-aware comparison with an indispensable KV-only control, and (iv) analytical limits on claims of novelty and linear-time operation. We do not claim to have trained or benchmarked the published Kimi Linear checkpoint.

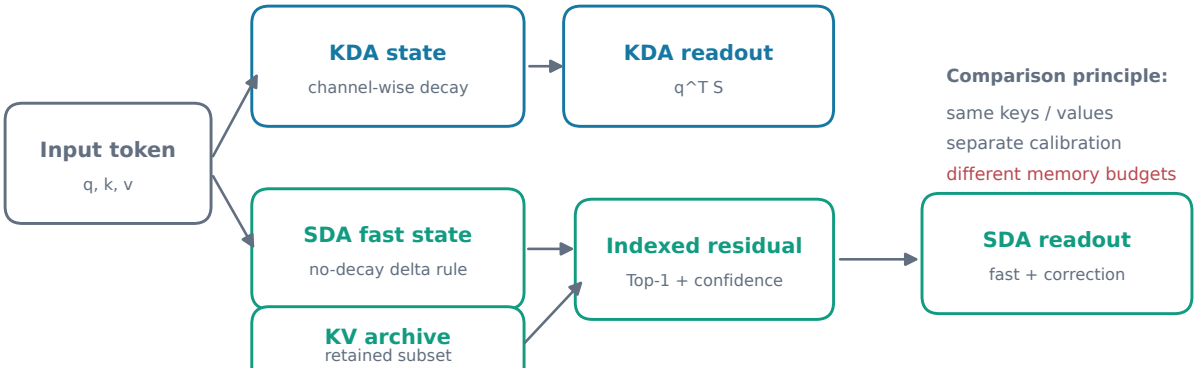


Figure 1: Comparison of memory pathways. KDA applies channel-wise decay and delta correction to a fixed-size state; SDA-5 retains a no-decay fast state plus a selected KV archive. The difference in storage capacity is fundamental to interpreting recall results.

2 Methods

2.1 KDA recurrence and scope of implementation

Let $S_t \in \mathbb{R}^{d_k \times d_v}$, $k_t, q_t \in \mathbb{R}^{d_k}$ and $v_t \in \mathbb{R}^{d_v}$. A KDA recurrence is [4]

$$\bar{S}_t = \text{Diag}(\boldsymbol{\alpha}_t) S_{t-1}, \tag{1}$$

$$S_t = \bar{S}_t + \beta_t k_t (v_t - \bar{S}_t^\top k_t)^\top, \tag{2}$$

$$o_t^{\text{KDA}} = S_t^\top q_t. \tag{3}$$

KDA permits learned, input-dependent $\boldsymbol{\alpha}_t \in (0,1]^{d_k}$. To isolate memory dynamics without training a large model, we evaluate a deterministic *channel-gated recurrence proxy* using

$$\alpha_{t,j} = \exp[-\ell c_j (1 + \eta \sqrt{d_k} |k_{t,j}|)], \quad c_j \in [0.5, 1.5]. \tag{4}$$

We select ℓ, η, β on development seeds. This family allows input-dependent channel-wise decay when $\eta > 0$, yet does not reproduce KDA's learned gate projections, short convolutions, multi-head organization, output gates, chunk kernels, or MLA hybridization. The best strictly positive-decay setting happens to select $\eta = 0$ here, so its final gate is channel-wise but time-invariant. We also report an unrestricted $\ell = 0$ boundary, which becomes no-decay DeltaNet.

2.2 SDA-5 indexed residual correction

The fast state $F_t \in \mathbb{R}^{d_k \times d_v}$ obeys

$$e_t = v_t - F_{t-1}^\top k_t, \tag{5}$$

$$F_t = F_{t-1} + \beta_t k_t e_t^\top. \tag{6}$$

An archive $\mathcal{M}_t$ stores B selected historical pairs (k_i, v_i). At query time, the nearest archived key is k_{i^*}, $i^* = \arg\max_{i \in \mathcal{M}_t} q^\top k_i$, and the binary confidence gate is $g(q) = \mathbf{1}[q^\top k_{i^*} \geq \tau]$. Its prediction is

$$o^{\text{SDA}}(q) = F_t^\top q + g(q)[v_{i^*} - F_t^\top k_{i^*}]. \tag{7}$$

For a missing or rejected match, SDA uses its recurrent predictor. The KV-only control returns v_{i^*} if accepted and zero otherwise (a strong synthetic-data baseline because values have mean zero). Archives are uniform random subsets, not learned selectors; retrieval uses *exact* nearest-neighbor scans rather than ANN.

Proposition 1 (Single-retrieval decomposition)**.** *For a retrieved pair (k_i, v_i) with $g(q) \in \{0, 1\}$, Equation (7) equals*

$$o^{\text{SDA}}(q) = F_t^\top (q - g k_i) + g v_i. \tag{8}$$

In particular, if $q = k_i$ and $g = 1$, the archived value is reconstructed exactly. If $g = 1$, the reconstruction error is bounded by $\|F_t\|_2 \|q - k_i\|_2$.

Proof. Expand and collect the two terms in $F_t^\top$; for $g = 1$ subtract v_i and apply the operator-norm inequality. □

This algebraic identity shows that retrieving a saved value is a crucial part of SDA's accuracy. Indexed KV retrieval is itself prior art [6]; the identity does not establish a new attention family.

2.3 Computational complexity

Each recurrent update costs $O(d_k d_v)$ and stores $O(d_k d_v)$ scalars for both recurrent mechanisms. SDA stores $O(B(d_k + d_v))$ additional KV scalars and, with the exact archive scan used here, performs $O(B d_k)$ similarity computation per query. For fixed d_k, d_v and $B \propto T$, repeated exact scans can yield quadratic total retrieval work in T. No claim of sublinear indexed retrieval or superior GPU throughput is made.

3 Experimental Protocol

We generate $N \in \{128, 512, 2048\}$ independent standard-normal keys in $\mathbb{R}^{32}$, each L2-normalized, and independent standard-normal values in $\mathbb{R}^{16}$. After writing all N pairs, 128 queries sample source indices with replacement. For source key k_j, the query is

$$q = \frac{k_j + \sigma z/\sqrt{32}}{\|k_j + \sigma z/\sqrt{32}\|_2}, \quad z \sim \mathcal{N}(0, I), \quad \sigma \in \{0, 0.4, 1\}. \tag{9}$$

The target is the original v_j. We evaluate MSE $= \frac{1}{128 d_v} \sum_{j=1}^{128} \|\hat{v}_j - v_j\|^2$. Dev seeds 20250001–20250008 select hyperparameters at $N = 512$, $\sigma \in \{0, 0.4\}$; test seeds 20261023–20261038 are independent and used for all reported results. All methods receive identical test keys, values, sources and query-noise realizations. For archive budgets $\rho \in \{0.1, 0.25, 0.5, 1\}$, the first round$(\rho N)$ positions of a fixed per-seed random permutation are archived. This produces nested archives across budgets.

The KDA proxy searches $\beta \in \{0.05, 0.1, 0.2, 0.4, 0.7, 1\}$, $\ell \in \{0, 0.0005, 0.002, 0.008, 0.03\}$ and $\eta \in \{0, 1\}$; we separately select the best setting constrained to $\ell > 0$. SDA searches the same β grid and $\tau \in \{0.65, 0.75, 0.8, 0.85, 0.9\}$ using 50% retention. The dev-selected SDA setting is $\beta = 0.05, \tau = 0.75$; the strictly positive KDA proxy uses $\beta = 0.05, \ell = 0.0005, \eta = 0$; the unrestricted KDA optimum uses $\ell = 0$. A prior SDA setting ($\beta = 1, \tau = 0.8$) is reported separately. A sharp full-KV reference uses softmax$(32qK^\top)V$: this is *not* an equal-memory or trained Transformer control. Computation is CPU NumPy float64; the reported KiB cover only stored state and KV arrays, not index metadata, activations, or model weights.

4 Results

4.1 Primary comparison

Table 1: Held-out mean value-MSE ± standard error over 16 seeds; $N = 512$, $\sigma = 0$, $\rho = 0.5$ for archive methods. Lower MSE is better.

Method	MSE ± SEM	State + KV (KiB)
Full KV softmax (sharp, all KV)	≈ 0	192
KDA channel-gated recurrence proxy	0.9336 ± 0.0098	**4**
KDA no-decay boundary	0.9301 ± 0.0098	**4**
DeltaNet, calibrated	0.9301 ± 0.0098	**4**
SDA-5 original settings	0.9422 ± 0.0221	100
SDA-5 calibrated	$\mathbf{0.4721 \pm 0.0099}$	100
Indexed KV, zero fallback	0.4962 ± 0.0101	96

The calibrated SDA-5 reduces MSE relative to the positive-decay KDA proxy by 0.4615 (paired 95% Student-t interval $[0.4384, 0.4845]$ in favor of SDA), while using *25 times* as much stored state/KV memory. It also improves modestly over the synthetic KV-only control by 0.0241 (paired 95% interval $[0.0217, 0.0265]$), with 4 KiB extra memory. These confidence intervals measure variation across the 16 chosen test seeds, not generalization to natural-language problems.

4.2 Retention and memory budget

At $N = 512$, zero noise, SDA-5 calibrated MSE equals $0.8475, 0.7098, 0.4721$ at 10%, 25%, and 50% retention; with all KVs retained it approaches numerical zero. This improvement accompanies increasing memory: 100 KiB at 50% retention versus 4 KiB for KDA. Figure 3 shows both reconstruction error and storage usage, including the strong indexed KV-only control.

4.3 Noise and length

At $\rho = 0.5$ and $N = 512$, query noise $\sigma = 1$ raises the calibrated SDA MSE to 0.8077, versus 0.9504 for the KDA proxy; the accepted-match fraction drops to 0.1484. For $N = 2048$ with exact queries, SDA-5 obtains 0.5207 MSE at

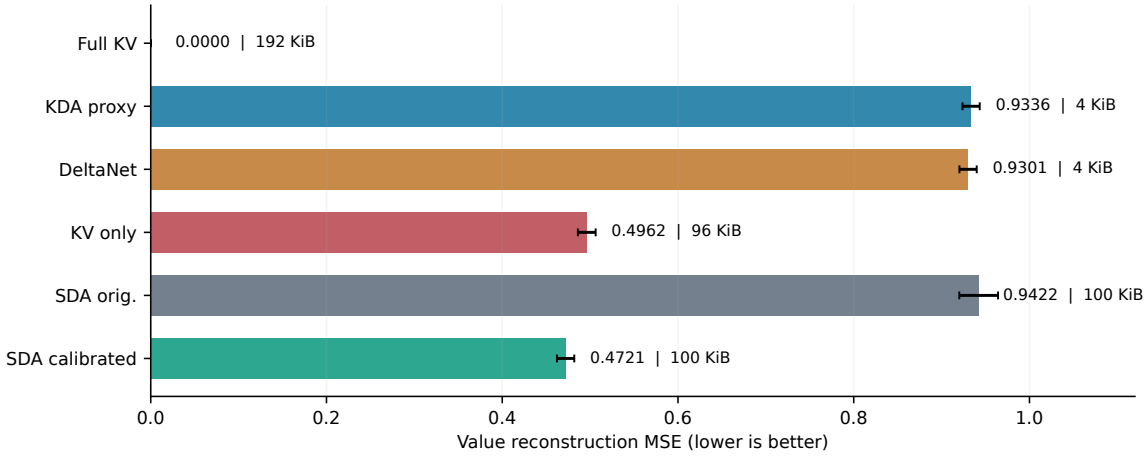


Figure 2: Primary comparison, with analytical state/KV memory shown after each MSE estimate. This is a cross-budget accuracy comparison, not an equal-memory contest.

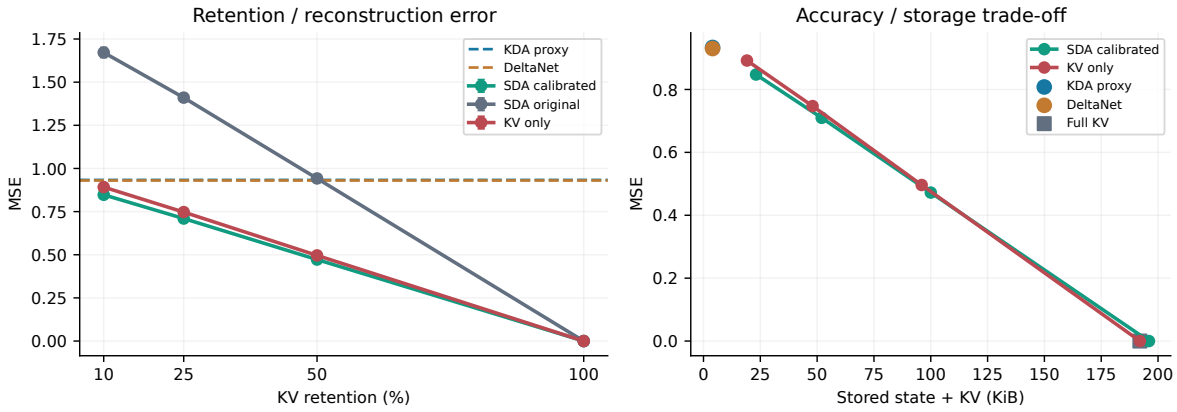


Figure 3: Left: MSE across archive budgets. Right: measured MSE versus analytical state+KV storage. KDA and DeltaNet use 4 KiB independent of sequence length; actual runtime and memory-allocation overheads are not included.

388 KiB versus KDA's 1.0105 at 4 KiB. Because hyperparameters were calibrated at $N = 512$, the longer-sequence experiment probes transfer of fixed settings rather than retuned optimal performance.

5 Discussion and Limitations

A fundamental capacity mismatch. KDA has no explicit growing KV archive, whereas SDA gains much of its exact recall from storing $B = \rho N$ selected original values. Better retrieval accuracy under these circumstances does not imply an intrinsically better memory-update rule; it demonstrates an accuracy/storage trade-off.

Not a comparison with the published Kimi model. The KDA baseline follows its state recurrence but uses a handcrafted, development-calibrated channel decay schedule, not trained learned gates, convolutional preprocessing, learned QKV projections, output gating, chunkwise kernels, or the Kimi Linear hybrid containing MLA. In this stationary random-association experiment, development calibration favors almost no forgetting. This is an observation about this narrow task, not evidence that channel-specific forgetting is useless for language modeling.

Other validity constraints. We evaluate after all writes, not per-token causal language modeling; random keys/values lack compositional semantics, and there are no changing values, collisions deliberately induced between sources, trained model weights, or GPU throughput measurements. A sharp Softmax scale of 32 makes full KV essentially a nearest-neighbor oracle. The confidence threshold is tuned on development seeds at a fixed 50% archive budget and does not establish robust calibration under shifts. Exact archive scanning is not linear-time in the number of stored entries. The 16 seeds quantify only synthetic sampling variation, not deployment uncertainty.

Novelty and related work. A hybrid of a delta-rule state and sparse KV retrieval overlaps with fast-weight memories [2] and Memorizing Transformers [6]. KDA, Gated DeltaNet-2 and Titans offer related alternative mechanisms [4, 5, 7]. Strong novelty requires demonstrably better retrieval or interference bounds, trained end-to-end integration,

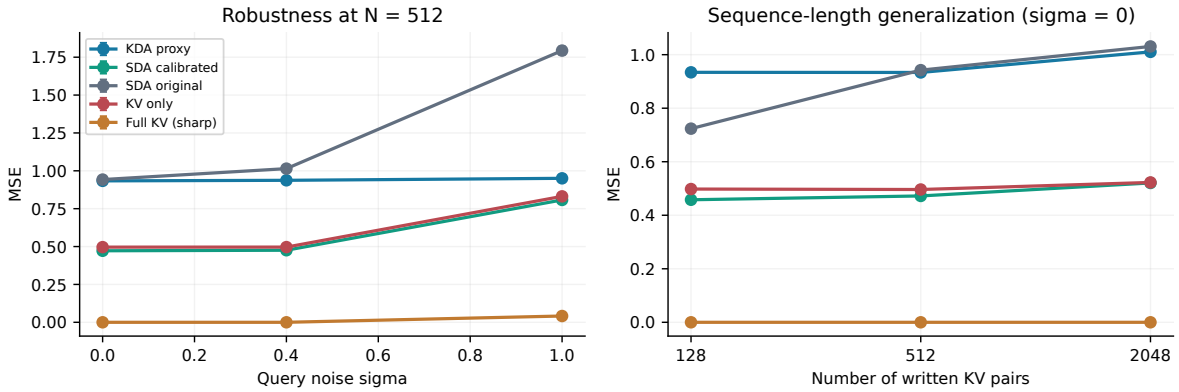


Figure 4: Noise robustness (left) and sequence-length transfer (right). Both panels compare the recurrent KDA proxy, calibrated SDA, uncalibrated SDA, indexed KV control and sharp full-KV reference.

and equal-memory efficiency improvements. None is yet established.

6 Conclusion

On a controlled synthetic exact-association task, calibrated SDA-5 achieves lower recall error than an untrained channel-gated KDA recurrence proxy. This comes with a substantial archive cost, and does not establish superiority over trained KDA/Kimi Linear, nor a linear-time or fixed-memory advantage. Future work should compare learned blocks at equal parameter count, FLOPs and total peak memory; learn gates and archives under causal token-by-token evaluation; measure long-context retrieval/perplexity and GPU latency; and include ANN-index overhead and simple KV controls.

A Implementation and Audit Trail

The accompanying `experiment.py` creates `configuration.json`, `raw_results.csv`, and `summary_results.csv`. `figures.py` recreates all figures. Execute `OPENBLAS_NUM_THREADS=1 python experiment.py` and then `python figures.py`. Deterministic RNG: `numpy.random.default_rng(seed)`. All results use float64. The script implements $\bar{S} = \mathrm{Diag}(\alpha)S$ followed by $S \leftarrow \bar{S} + \beta k(v - \bar{S}^\top k)^\top$ without materializing the $d_k \times d_k$ transition. The test split is never used during hyperparameter selection. The original SDA setting is included explicitly to prevent hiding sensitivity to β and the threshold.

References

[1] Vaswani, A. et al. (2017). Attention Is All You Need. NeurIPS. https://arxiv.org/abs/1706.03762.
[2] Schlag, I., Irie, K., Schmidhuber, J. (2021). Linear Transformers Are Secretly Fast Weight Programmers. ICML. https://proceedings.mlr.press/v139/schlag21a.html.
[3] Yang, S., Kautz, J., Hatamizadeh, A. (2025). Gated Delta Networks: Improving Mamba2 with Delta Rule. ICLR. https://arxiv.org/abs/2412.06464.
[4] Kimi Team et al. (2025). Kimi Linear: An Expressive, Efficient Attention Architecture. arXiv:2510.26692. https://arxiv.org/abs/2510.26692.
[5] Hatamizadeh, A., Choi, Y., Kautz, J. (2026). Gated DeltaNet-2: Decoupling Erase and Write in Linear Attention. arXiv:2605.22791. https://arxiv.org/abs/2605.22791.
[6] Wu, Y., Rabe, M. N., Hutchins, D., Szegedy, C. (2022). Memorizing Transformers. ICLR. https://arxiv.org/abs/2203.08913.
[7] Behrouz, A., Zhong, P., Mirrokni, V. (2025). Titans: Learning to Memorize at Test Time. https://arxiv.org/abs/2501.00663.